

## G-bundles over G spaces

$G$  is a topological group, then by  $G$ -space we mean a topological space  $X$  together with a continuous action of  $G$  on  $X$ . A  $G$ -map is a map that respects the  $G$ -action between  $G$ -spaces.

A  $G$ -space  $E$  is a  $G$ -vector bundle over the  $G$ -space  $X$  if

1.  $E$  is a vector bundle over  $X$ .
2. the projection  $E \rightarrow X$  is a  $G$ -map.
3. for each  $g \in G$  the map  $E_x \rightarrow E_{gx}$  is a vector space homomorphism.

If  $X = \{x\}$  then a vector bundle over  $X$  is just a representation space of  $G$  over the fiber vector space.

$G$  is a finite group. Suppose  $X$  is a free(faithful)  $G$ -space. Let  $X/G$  be the orbit space. If  $E$  is a  $G$  vector bundle over the space  $X$ , then  $E$  is also a free  $G$  space. and  $E/G$  has a natural vector bundle structure over  $X/G$ . Conversely if  $F$  is a vector bundle over  $X/G$ , then  $q^*(F) \rightarrow X$  is a  $G$ -equivariant bundle map and  $q^*(F)_x \simeq q^*(F)_{gx}$ .

**Proposition 0.1.** *If  $X$  is a free  $G$ , then*

$$\{G - \text{vector bundles over } X\} \leftrightarrow \{\text{vector bundles over } \frac{X}{G}\}$$

by the map  $E \rightarrow E/G$ .

The natural map for any  $V$ ,

$$\sum V_i \otimes \text{hom}_G(V_i, V) \rightarrow V$$

is a  $G$ -isomorphism.

Define for a  $E$  a  $G$ -bundle over a trivial  $G$ -space  $X$ ,  $Av \in \text{END}(V)$  by

$$Av(e) = \frac{1}{|G|} \sum_{g \in G} g(e)$$

$\dim(\text{hom}^g(V, W))$   
is the  
multiplicity of  
 $W$  in  $V$ .

This is a projection operator and the image is a vector subbundle denoted by  $E^G$ , the invariant sub-bundle of  $E$ . So if  $E, F$  are two  $G$  bundles then  $\text{hom}_G(E, F) = \text{hom}(E, F)^G$  is again a vector bundle.

Taking  $E$  to be the trivial vector bundle  $V_i = X \times V_i$  we have a natural bundle map which is an isomorphism of  $G$ -bundles

$$\sum_{i=1}^k V_i \otimes \text{hom}_G(V_i, F) \rightarrow F$$

Thus every  $G$ -bundle  $G$  is isomorphic to a  $G$ -bundle of the form  $\sum V_i \otimes E_i$  where  $E_i$  is a vector bundle with trivial  $G$ -action, which is unique upto isomorphism ( $\simeq \text{hom}_G(V_i, F)$ ).

**Proposition 0.2.** *Let  $X$  be a trivial  $G$ -space, and  $V_1, \dots, V_k$  be a complete set of irreducible  $G$ -modules, which are trivial over  $X$ . Then every  $G$ -bundle  $F$  over  $X$  is isomorphic to a direct sum  $\sum V_i \otimes E_i$ , where the  $E_i \simeq \text{hom}_G(V_i, F)$  are unique upto isomorphism with trivial  $G$ -action.*

If  $E$  is a  $G$ -bundle then we have

$$\begin{aligned} G \times \Gamma(E) &\rightarrow \Gamma(E) \\ g, s &\mapsto g s g^{-1} \end{aligned}$$

The operator  $Av = \frac{1}{|G|} \sum_{g \in G} g$  is a projection operator from  $\Gamma(E) \rightarrow \Gamma(E)^G$ .

**Lemma 0.3.** *Let  $X$  be a compact  $G$ -space and  $Y \subset X$  be a closed sub  $G$ -space. Let  $E$  be a  $G$ -bundle over  $X$ . Then any invariant section  $s$  over  $Y$  extends to an invariant section over  $X$ .*

*Proof.* Extend  $s$  to a section  $t$  over  $X$ . Then  $Av(t)$  is an invariant section of  $E$  over  $X$  which is equal to  $s$  over  $Y$ .  $\square$

If  $E, F$  are two  $G$ -bundles then  $\text{hom}(E, F)$  is also  $G$ -bundle and we have  $\Gamma(\text{hom}(E, F))^G \simeq \text{HOM}_G(E, F)$

**Lemma 0.4.** *Let  $Y$  be a compact  $G$  space and  $X$  be a  $G$ -space.  $f_t : Y \rightarrow X$ ,  $G$  homotopy and  $E$  a  $G$  vector bundle over  $X$ . Then  $f_0^*(E)$  and  $f_1^*(E)$  are isomorphic  $G$  bundles*

A  $G$ -homotopy means a  $G$ -map  $F : Y \times I \rightarrow X$  where  $I$  is the unit interval with trivial  $G$  action (which is  $f_t(g \cdot y) = g f_t(y)$ ). A trivial  $G$ -bundle we mean a  $G$ -bundle isomorphic to a product  $X \times V$  where  $V$  is a  $G$ -module.  $h$  is a metric for  $E$  implies  $Av(h)$  is an invariant metric.

If  $V \subset \Gamma(E)$  is ample then  $\sum gV \subset \Gamma(E)$  is ample and invariant.

To classify  $G$ -bundles, we look at Grassmanians of  $G$ -subspaces of  $m \sum_{i=1}^k V_i = m[G]$  for  $m \rightarrow \infty$ .

$$\lim_{m \rightarrow \infty} [X, G_n(mR[G])]^G \simeq G \text{Vect}_n(X)$$

$A = C(X)$ . If  $X$  is a  $G$ -space,  $G$  acts on  $A$  as a group of algebra automorphisms.

If  $E$  is a  $G$ -vector bundle over  $X$ , then  $\Gamma(E)$  is a projective  $A$ -module, and  $G$  acts on  $\Gamma(E)$ ,

$$g(as) = g(a)g(s) \quad a \in A, g \in G, s \in \Gamma(E)$$

Consider the twisted group algebra  $B$  of  $G$  over  $A$ , elements of  $B$  are linear combinations  $\sum_{g \in G} a_g g$  with  $a_g \in A$  and the products is

$$(ag)(a'g') = (ag(a'))gg'$$

Then  $\Gamma(E)$  is a  $B$ -module. So  $\Gamma$  is a functor from  $G$ -vector bundles to finitely generated  $B$ -modules. If  $E$  is a trivial  $G$ -bundle, then  $\Gamma(E)$  is free of rank  $n$ .

So  $\Gamma$  is now a functor inducing equivalences between category of trivial vector bundles and category of free  $B$ -modules. Consider the projection operator on these two categories. This gives us an equivalence

$$\{G\text{-vector bundles}\} \leftrightarrow \{\text{finitely generated } B\text{-modules and projective over } A\}$$